

Pracownia Systemów i Urządzeń Sieciowych		
14.09.2022 - Konfiguracja urządzeń - ruting, dhcp i VLANy		
	Imię i Nazwisko	klasa/grupa/ rok szk.
	Ksawery Zelek	3TI 4/4

TREŚĆ ZADANIA

Konfiguracja routera:

- Serwer DHCP .60 -> .80
- Rezerwacja adresów:

PC1 -> 192.168.0.62

PC2 -> 192.168.0.72

- Uruchomiona sieć WiFi:

SSID -> Numer Stanowiska

5GHz -> Wyłączone

- Hasło: zst1234!

1. PINGI PO PODŁĄCZENIU URZĄDZEŃ

STACJA ROBOCZA

```
Administrator: Wiersz polecenia
Microsoft Windows [Version 10.0.19044.1766]
(c) Microsoft Corporation. Wszelkie prawa zastrzeżone.

C:\WINDOWS\system32>ping 192.168.0.10

Pinging 192.168.0.10 with 32 bytes of data:
Reply from 192.168.0.10: bytes=32 time=1ms TTL=64
Reply from 192.168.0.10: bytes=32 time=1ms TTL=64
Reply from 192.168.0.10: bytes=32 time=1ms TTL=64
Reply from 192.168.0.10: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\WINDOWS\system32>ping 192.168.0.20

Pinging 192.168.0.20 with 32 bytes of data:
Reply from 192.168.0.20: bytes=32 time=3ms TTL=64
Reply from 192.168.0.20: bytes=32 time=1ms TTL=64
Reply from 192.168.0.20: bytes=32 time=1ms TTL=64
Reply from 192.168.0.20: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\WINDOWS\system32>ping 192.168.0.30

Pinging 192.168.0.30 with 32 bytes of data:
Reply from 192.168.0.30: bytes=32 time=4ms TTL=64
Reply from 192.168.0.30: bytes=32 time=1ms TTL=64
Reply from 192.168.0.30: bytes=32 time=1ms TTL=64
Reply from 192.168.0.30: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.0.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 1ms

C:\WINDOWS\system32>ping 192.168.0.50

Pinging 192.168.0.50 with 32 bytes of data:
Reply from 192.168.0.64: Destination host unreachable.
Reply from 192.168.0.40: Destination host unreachable.
Reply from 192.168.0.64: Destination host unreachable.
Reply from 192.168.0.64: Destination host unreachable.

Ping statistics for 192.168.0.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

C:\WINDOWS\system32>
```

SERWER

```
Administrator: Wiersz polecenia
Microsoft Windows [Version 10.0.19044.1766]
(c) Microsoft Corporation. Wszelkie prawa zastrzeżone.

C:\WINDOWS\system32>ping 192.168.0.10

Pinging 192.168.0.10 with 32 bytes of data:
Reply from 192.168.0.10: bytes=32 time=1ms TTL=64
Reply from 192.168.0.10: bytes=32 time=1ms TTL=64
Reply from 192.168.0.10: bytes=32 time=1ms TTL=64
Reply from 192.168.0.10: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\WINDOWS\system32>ping 192.168.0.20

Pinging 192.168.0.20 with 32 bytes of data:
Reply from 192.168.0.20: bytes=32 time=3ms TTL=64
Reply from 192.168.0.20: bytes=32 time=2ms TTL=64
Reply from 192.168.0.20: bytes=32 time=1ms TTL=64
Reply from 192.168.0.20: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\WINDOWS\system32>ping 192.168.0.30

Pinging 192.168.0.30 with 32 bytes of data:
Reply from 192.168.0.30: bytes=32 time=4ms TTL=64
Reply from 192.168.0.30: bytes=32 time=1ms TTL=64
Reply from 192.168.0.30: bytes=32 time=1ms TTL=64
Reply from 192.168.0.30: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.0.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 1ms

C:\WINDOWS\system32>ping 192.168.0.40

Pinging 192.168.0.40 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.0.40:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\WINDOWS\system32>
```

2. USTAWIENIE ZAKRESU SERWERA DHCP + REZERWACJA ADRESÓW

STACJA ROBOCZA

```
Administrator: Wiersz polecenia
C:\WINDOWS\system32\ipconfig /all

Windows IP Configuration

Host Name . . . . . : 24p-00
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Ethernet Connection (2) I219-V
Physical Address. . . . . : 4C-CC-6A-D7-DA-60
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . : Yes
Link-local IPv6 Address . . . . : fe80::887b:67d7:a951:290c%15(Preferred)
IPv4 Address. . . . . : 192.168.0.62(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : czwartek, 15 września 2022 12:47:49
Lease Expires . . . . . : niedziela, 22 października 2158 19:16:18
Default Gateway . . . . . : 192.168.0.10
DHCP Server . . . . . : 192.168.0.10
DHCPv6 IAID . . . . . : 38587498
DHCPv6 Client DUID. . . . . : 00-01-00-01-2A-B4-7D-4A-4C-CC-6A-D7-DA-60
DNS Servers . . . . . : 192.168.0.10
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet 3:

Connection-specific DNS Suffix . :
Description . . . . . : TP-Link Gigabit PCI Express Adapter
Physical Address. . . . . : 18-06-C7-04-48-05
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . : Yes
Link-local IPv6 Address . . . . : fe80::d486:d76:8432:85d%13(Preferred)
IPv4 Address. . . . . : 192.168.0.48(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.0.10
DHCPv6 IAID . . . . . : 511478730
DHCPv6 Client DUID. . . . . : 00-01-00-01-2A-B4-7D-4A-4C-CC-6A-D7-DA-60
DNS Servers . . . . . : 192.168.0.10
NetBIOS over Tcpip. . . . . : Enabled

C:\WINDOWS\system32_
```

SERWER

```
Administrator: Wiersz polecenia
C:\WINDOWS\system32\ipconfig /all

Windows IP Configuration

Host Name . . . . . : 24p-00
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Ethernet Connection (2) I219-V
Physical Address. . . . . : 4C-CC-6A-F5-A1-5D
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . : Yes
Link-local IPv6 Address . . . . : fe80::d9a2:38ea:1aae:2c46%15(Preferred)
IPv4 Address. . . . . : 192.168.0.77(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : czwartek, 15 września 2022 12:47:31
Lease Expires . . . . . : niedziela, 22 października 2158 19:17:09
Default Gateway . . . . . : 192.168.0.10
DHCP Server . . . . . : 192.168.0.10
DHCPv6 IAID . . . . . : 38587498
DHCPv6 Client DUID. . . . . : 00-01-00-01-2A-B4-61-21-4C-CC-6A-F5-A1-5D
DNS Servers . . . . . : 192.168.0.10
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet 3:

Connection-specific DNS Suffix . :
Description . . . . . : TP-Link Gigabit PCI Express Adapter
Physical Address. . . . . : CA-dE-1F-4a-3a-02
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . : Yes
Link-local IPv6 Address . . . . : fe80::4c1a:f79b:1e08:eb2%13(Preferred)
IPv4 Address. . . . . : 192.168.0.58(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.0.10
DHCPv6 IAID . . . . . : 511478730
DHCPv6 Client DUID. . . . . : 00-01-00-01-2A-B4-61-21-4C-CC-6A-F5-A1-5D
DNS Servers . . . . . : 192.168.0.10
NetBIOS over Tcpip. . . . . : Enabled

C:\WINDOWS\system32_
```

KONFIGURACJA NA ROUTERZE -> DHCP

The screenshot displays the TP-Link Archer C1200 web interface in Polish. The browser address bar shows the URL `192.168.0.10/webpages/index.1516243669548.html`. The interface has a teal header with the TP-Link logo and navigation tabs: 'Szybka konfiguracja', 'Podstawowe', and 'Zaawansowane'. The 'Zaawansowane' tab is selected, and the 'Sieć' (Network) section is highlighted in the left sidebar. The 'Ustawienia' (Settings) page for DHCP is shown, with the 'Włącz serwer DHCP' checkbox checked. The IP range is set to 192.168.0.60 - 192.168.0.80, and the lease time is 120 minutes. The default gateway is 192.168.0.10. Below the settings, there is a 'Rezerwacja adresów' (IP Reservation) section with a table of reserved addresses. At the bottom, a 'Lista klientów DHCP' (DHCP Clients) section shows 3 clients. The footer of the interface provides version information: 'Wersja firmware: 2.0.3 Build 20211219 Rel. 67401 (EU)' and 'Wersja sprzętowa: Archer C1200 v2.0'.

Wireless Router Archer C1200

Niezabezpieczona | 192.168.0.10/webpages/index.1516243669548.html

tp-link

Stan

Sieć

- Internet
- LAN
- IPTV/VLAN
- Serwer DHCP
- Dynamiczny DNS
- Routing zaawansowany

Tryb pracy

Sieć bezprzewodowa

Sieć dla gości

Przekierowanie portów

Ustawienia USB

Kontrola rodzicielska

QoS

Szybka konfiguracja Podstawowe Zaawansowane

Polski

Dioda Wyloguj Resetuj

Ustawienia

Serwer DHCP: ☒ Włącz serwer DHCP

Pula adresów IP: 192.168.0.60 - 192.168.0.80

Czas przydzielenia adresu: 120 minut. (1-2880. Domyślna wartość to 120.)

Brama domyślna: 192.168.0.10 (Opcjonalnie)

Preferowany DNS: (Opcjonalnie)

Alternatywny DNS: (Opcjonalnie)

Zapisz

Rezerwacja adresów

[Dodaj](#) [Kasuj](#)

☐	ID	Adres MAC	Zarezerwowany adres IP	Opis	Stan	Zmień
☐	1	4C-CC-6A-D7-DA-60	192.168.0.62	---	🔧	✎ 🗑
☐	2	4C-CC-6A-F5-A1-5D	192.168.0.72	---	🔧	✎ 🗑

Lista klientów DHCP

Ilość klientów: 3 [Odśwież](#)

ID	Nazwa klienta	Adres MAC	Przydzielone IP	Czas przydzielenia
----	---------------	-----------	-----------------	--------------------

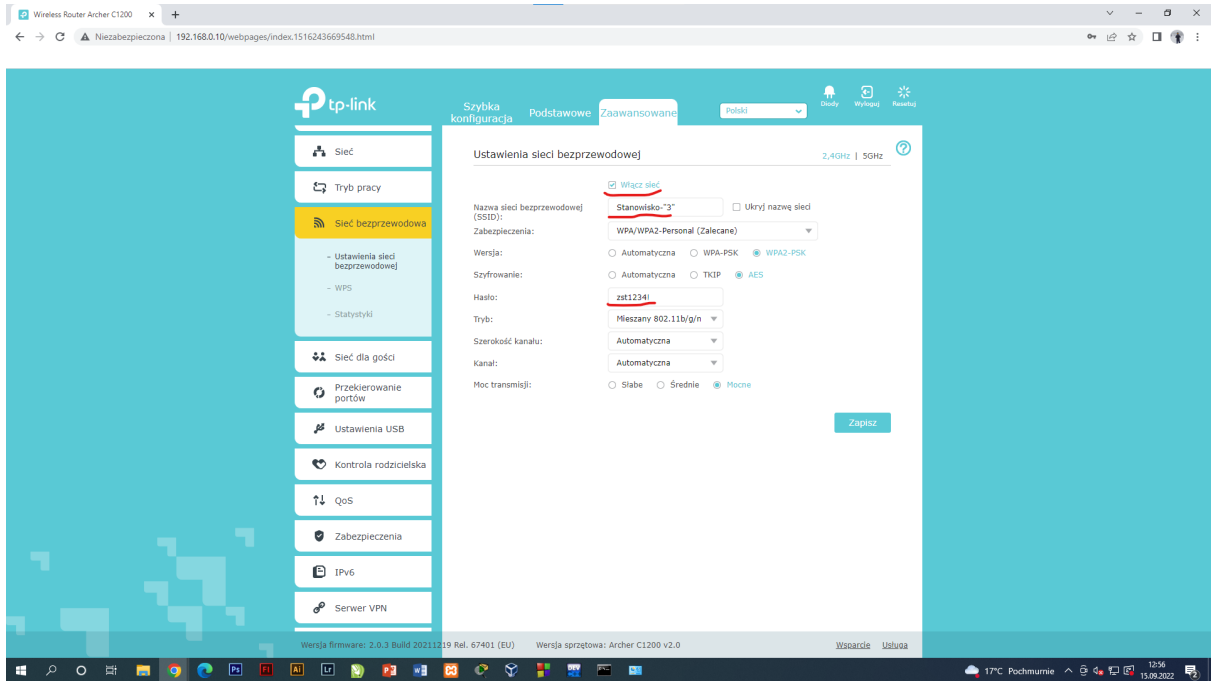
Wersja firmware: 2.0.3 Build 20211219 Rel. 67401 (EU) Wersja sprzętowa: Archer C1200 v2.0

Wspomóż Usługa

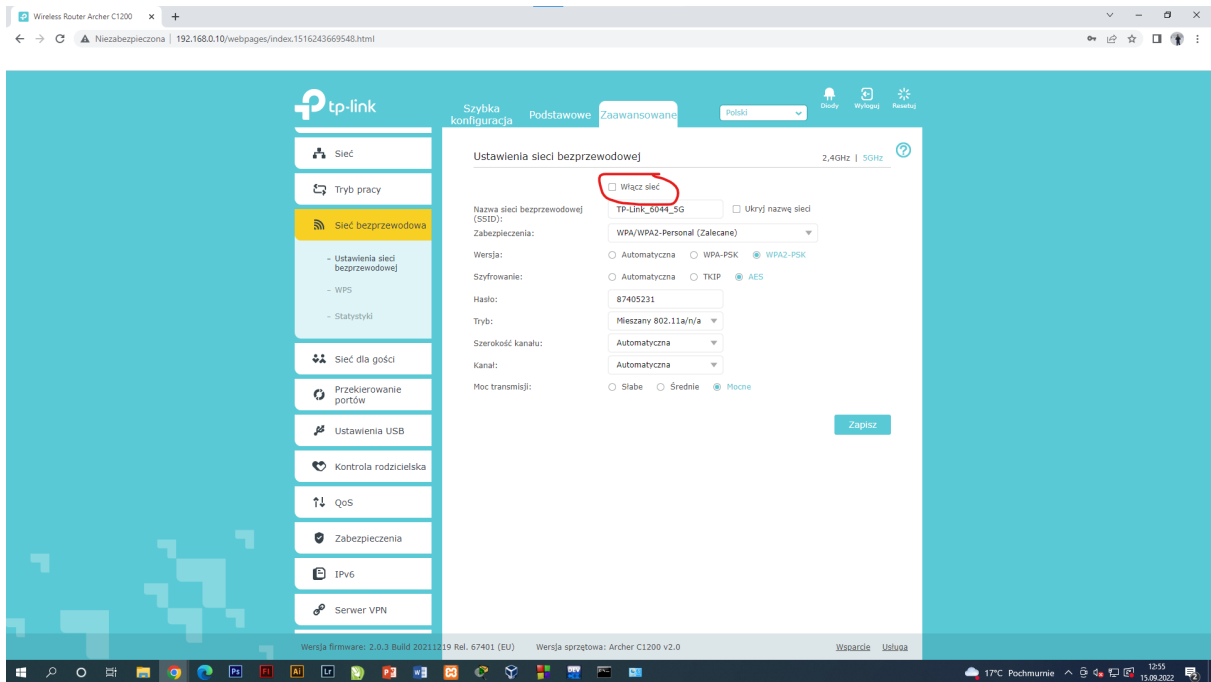
17°C Pochmurnie 12:50 13.09.2022

3. URUCHOMIENIE WIFI

SSID



SIEĆ 5GHz



4. VLANy i PVID

tp-link T1600-28TS

TP-Link T1600G-28TS

VLAN Config | Port Config

System
Switching
VLAN
• 802.1Q VLAN
• MAC VLAN
• Protocol VLAN
Spanning Tree
Multicast
Routing
QoS
ACL
Network Security
SNMP
LLDP
Maintenance
Save Config
Index
Logout

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Select	VLAN_ID	Name	Members	Operation
☐	1	System-VLAN	1/0/1-28	Edit Detail
☐	10	VLAN10	1/0/1, 1/0/3, 1/0/6	Edit Detail
☐	20	VLAN20	1/0/3, 1/0/6, 1/0/8	Edit Detail

[All](#) [Create](#) [Delete](#) [Help](#)

Total VLAN: 3

TP-Link T1600G-28TS

VLAN Config | Port Config

System
Switching
VLAN
• 802.1Q VLAN
• MAC VLAN
• Protocol VLAN
Spanning Tree
Multicast
Routing
QoS
ACL
Network Security
SNMP
LLDP
Maintenance
Save Config
Index
Logout

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Select	UNIT	Port	PVID	LAG	VLAN
☐	1	1/0/1	10	---	Detail
☐	1	1/0/2	1	---	Detail
☐	1	1/0/3	1	---	Detail
☐	1	1/0/4	1	---	Detail
☐	1	1/0/5	1	---	Detail
☐	1	1/0/6	20	---	Detail
☐	1	1/0/7	1	---	Detail
☐	1	1/0/8	1	---	Detail
☐	1	1/0/9	1	---	Detail
☐	1	1/0/10	1	---	Detail
☐	1	1/0/11	1	---	Detail
☐	1	1/0/12	1	---	Detail
☐	1	1/0/13	1	---	Detail
☐	1	1/0/14	1	---	Detail
☐	1	1/0/15	1	---	Detail

[All](#) [Apply](#) [Help](#)

TL-SG105E

TL-SG105E

System
Switching
Monitoring
VLAN
• MTU VLAN
• Port Based VLAN
• 802.1Q VLAN
• 802.1Q PVID Setting
QoS
Save Config
Logout

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802.1Q VLAN Configuration

802.1Q VLAN Configuration: ☒ Enable ☐ Disable [Apply](#)

VLAN ID	Port	Untagged	Tagged	Not Member
1	Port 1	☐	☐	☒
10	Port 2	☐	☐	☒
20	Port 3	☐	☐	☒
	Port 4	☐	☐	☒
	Port 5	☐	☐	☒

[Add/Modify](#) [Help](#)

VLAN ID	VLAN Name	Member Ports	Tagged Ports	Untagged Ports	Delete
1	Default_VLAN	1-5		1-5	☐
10	VLAN10	1,3	3	1	☐
20	VLAN20	3,5	3	5	☐

[Select All](#) [Delete](#)

TL-SG105E

System
Switching
Monitoring
VLAN
• MTU VLAN
• Port Based VLAN
• 802.1Q VLAN
• 802.1Q PVID Setting
QoS
Save Config
Logout

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802.1Q VLAN PVID Setting

Select	Port	PVID
☐	Port 1	10
☐	Port 2	1
☐	Port 3	1
☐	Port 4	1
☐	Port 5	1

[Apply](#) [Help](#)

Note:
1. By default, the PVID of all ports is 1.
2. 802.1Q VLAN PVID will be restored to 1 when 802.1Q VLAN is disabled.

PINGI

```
C:\WINDOWS\system32>ping 192.168.0.62
```

```
Pinging 192.168.0.62 with 32 bytes of data:  
Reply from 192.168.0.62: bytes=32 time<1ms TTL=128  
Reply from 192.168.0.62: bytes=32 time<1ms TTL=128  
Reply from 192.168.0.62: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.0.62:  
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 0ms, Maximum = 0ms, Average = 0ms  
Control-C  
^C
```

```
C:\WINDOWS\system32>ping 192.168.0.40
```

```
Pinging 192.168.0.40 with 32 bytes of data:  
Request timed out.  
Request timed out. _____  
Reply from 192.168.0.40: bytes=32 time<1ms TTL=128  
Reply from 192.168.0.40: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.0.40:  
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

odpięte

podpięte

```
C:\WINDOWS\system32>ping 192.168.0.72
```

```
Pinging 192.168.0.72 with 32 bytes of data:  
Reply from 192.168.0.72: bytes=32 time<1ms TTL=128  
Reply from 192.168.0.72: bytes=32 time<1ms TTL=128  
Reply from 192.168.0.72: bytes=32 time<1ms TTL=128  
Reply from 192.168.0.72: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.0.72:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\WINDOWS\system32>ping 192.168.0.50
```

```
Pinging 192.168.0.50 with 32 bytes of data:  
Reply from 192.168.0.62: Destination host unreachable.  
Reply from 192.168.0.62: Destination host unreachable.  
Reply from 192.168.0.50: bytes=32 time<1ms TTL=128  
Reply from 192.168.0.50: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.0.50:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

odpięte

podpięte